

WASP-12b

R-2099

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150125 · created 2026-08-10T23:13:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457048.71173 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.044
Transit depth (Rp/Rs)^2	2.37 +/- 1.35 [%]
Orbital Inclination (inc)	86.27 +/- 2.86
Airmass coefficient 1 (a1)	1.39 +/- 0.17
Airmass coefficient 2 (a2)	-0.29 +/- 0.21
Scatter in the residuals of the lightcurve fit is	12.41 %
Best Comparison Star	#3 - [527, 93]
Optimal Aperture	2.66
Optimal Annulus	7.28
Transit Duration (day)	0.132 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.044 vs 0.1178 (deviation 0.78 σ)
Duration fit vs published	0.132 d vs 0.125 d (deviation 0.98 σ)
Mid-time O-C	1571.9 min
Depth SNR	3.3
Residual scatter	12.41 %

Light curve

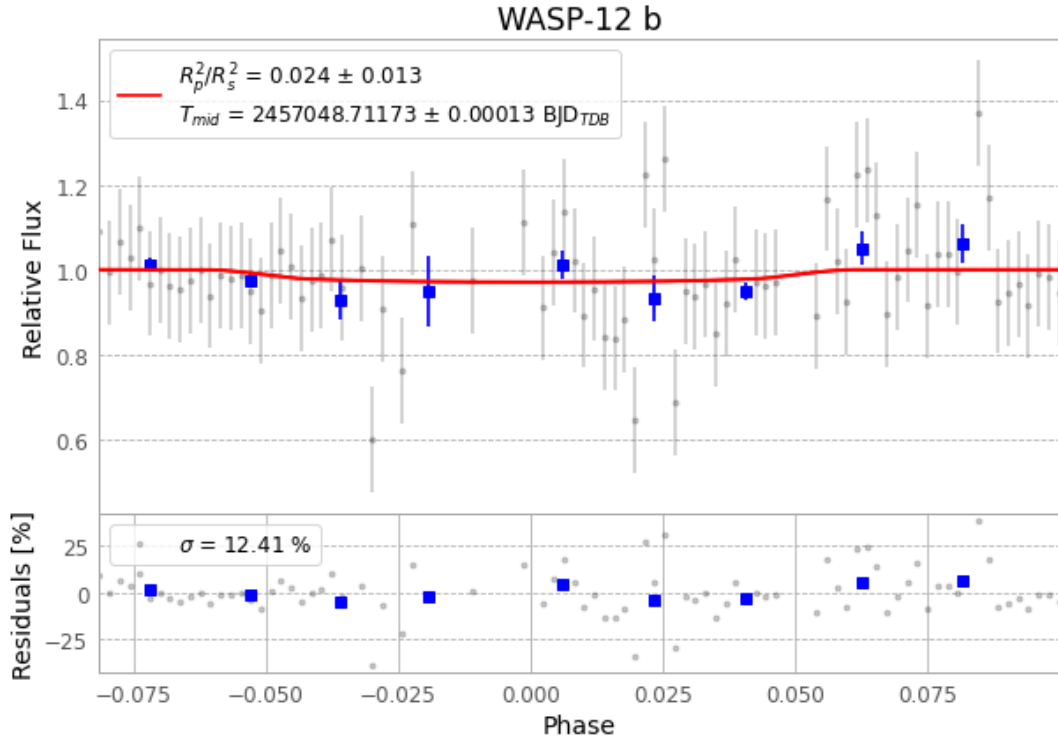


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-25.png (SHA-256 7d427ed0b0a2b56b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 adb1c75757a29bb3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

98 FITS frames (64 MB), WASP-12150126024812.FITS → WASP-12150126073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150125.log (c32ce1384f04...), FinalLightCurve_WASP-12 b_2015-01-25.png (7d427ed0b0a2...), FinalLightCurve_WASP-12 b_2015-01-25.csv (28e9f60d36b4...)

Compute resources

Wall time 253.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:14Z.